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IMPLEMENTATION COMPLETION AND RESULTS REPORT
ON A
IDA CREDIT IN THE AMOUNT OF SDR1.5 MILLION
(US\$2.0 MILLION EQUIVALENT)

IDA GRANT IN THE AMOUNT OF SDR1.5 MILLION
(US\$2.0 MILLION EQUIVALENT)

SCF - SREP GRANT IN THE AMOUNT OF
US\$6.77 MILLION

PRIF GRANT IN THE AMOUNT OF
USD\$3.4 MILLION

TO THE
REPUBLIC OF VANUATU

FOR THE
RURAL ELECTRIFICATION PROJECT STAGE II (P160658)
December 05, 2022

Energy and Extractives Global Practice
East Asia and Pacific Region

CURRENCY EQUIVALENTS

(Exchange Rate Effective June 30, 2022)

Currency Unit = Vanuatu Vatu (VUV)

117.08 VUV = US\$1

US\$1.32778 = SDR 1

FISCAL YEAR

July 1 - June 30

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ABBREVIATIONS AND ACRONYMS

CIF	Climate Investment Funds
COM	Council of Ministers (Vanuatu)
DC	Direct Current
DoE	Department of Energy (Vanuatu)
ECOP	Environmental Code of Practice
EIRR	Economic Rate of Return
ESIA	Environmental and Social Impact Assessment
ESMF	Environmental and Social Management Framework
ESMAP	Energy Sector Management Assistance Program
ESMP	Environmental and Social Management Plan
FIRR	Financial Internal Rate of Return
GDP	Gross Domestic Product
GHG	Greenhouse Gases
GoV	Government of Vanuatu
GPOBA	Global Partnership for Output-Based Aid
HIES	Household Income and Expenditure Survey
IFRs	Interim Financial Reports
INDC	Intended Nationally Determined Contribution
kW	Kilowatts
kWh	Kilowatt-hours
LCOE	Levelized Cost of Energy
MW	Megawatt
MWh	Megawatt-hours
NPV	Net Present Value
NERM	National Energy Road Map (Vanuatu)
NZMFAT	New Zealand Ministry of Foreign Affairs and Trade
OVR	Output Verification Report
PIC	Pacific Island Countries
POM	Project Operations Manual
PRIF	Pacific Region Infrastructure Facility
PV	Photovoltaic
RPF	Regional Partnership Framework
SCD	Systematic Country Dialogue
SE4All	Sustainable Energy for All
SHS	Solar Home System
SIA	Subsidy Implementation Agreement
SIM	Subsidy Implementation Manual
SREP	Scaling-Up Renewable Energy Program
UNELCO	<u>Union Electrique de Vanuatu Ltd</u>
URA	Utilities Regulatory Authority (Vanuatu)
VREP I	Vanuatu Rural Electrification Project Stage I
VREP II	Vanuatu Rural Electrification Project Stage II
VUI	Vanuatu Utilities and Infrastructure
VUV	Vanuatu Vatu
WBG	World Bank Group
Wp	Watt Peak
WTP	Willingness to Pay

TABLE OF CONTENTS

DATA SHEET	1
I. PROJECT CONTEXT AND DEVELOPMENT OBJECTIVES.....	5
A. CONTEXT AT APPRAISAL	5
B. SIGNIFICANT CHANGES DURING IMPLEMENTATION	11
II. OUTCOME	12
A. RELEVANCE TO PDOs	12
B. ACHEIVEMENT OF PDOs (EFFICACY)	13
III. KEY FACTORS THAT AFFECTED IMPLMENTATION AND OUTCOME.....	20
A. KEY FACTORS DURING PREPARATION	20
B. KEY FACTORS DURING IMPLMENTATION.....	20
IV. BANK PERFORMANCE, COMPLIANCE ISSUES, AND RISK TO DEVELOPMENT OUTCOME ..	22
A. QUALITY OF MONITORING AND EVALUATION (M&E)	22
B. ENVIRONMENTAL, SOCIAL AND FIDUCIARY COMPLIANCE.....	23
C. BANK PERFORMANCE	24
D. RISK TO DEVELOPMENT OUTCOME	25
V. LESSONS AND RECOMMENDATIONS	26
ANNEX 1. RESULTS FRAMEWORK AND KEY OUTPUTS.....	28
ANNEX 2. BANK LENDING AND IMPLEMENTATION SUPPORT/SUPERVISION	35
ANNEX 3. PROJECT COST BY COMPONENT	37
ANNEX 4. EFFICIENCY ANALYSIS.....	38
ANNEX 5. BORROWER, CO-FINANCIER AND OTHER PARTNER/STAKEHOLDER COMMENTS ...	39
ANNEX 6. SUPPORTING DOCUMENTS (IF ANY)	40



DATA SHEET

BASIC INFORMATION

Product Information

Project ID	Project Name
P160658	Rural Electrification Project Stage II
Country	Financing Instrument
Vanuatu	Investment Project Financing
Original EA Category	Revised EA Category
Partial Assessment (B)	Partial Assessment (B)

Organizations

Borrower	Implementing Agency
Ministry of Finance and Economic Management	Department of Energy, Ministry is Ministry of Climate Change Adaptation, Meteorology & Geohazards, Environment, Energy and

Project Development Objective (PDO)

Original PDO

The Project Development Objective is to support increased penetration of renewable energy and increased access to electricity services in the dispersed off-grid areas of Vanuatu.

**FINANCING**

	Original Amount (US\$)	Revised Amount (US\$)	Actual Disbursed (US\$)
World Bank Financing			
TF-A5406	3,450,000	3,450,000	1,017,517
TF-A4979	6,770,000	6,770,000	580,680
IDA-60720	2,000,000	2,000,000	190,348
IDA-D1930	2,000,000	2,000,000	490,842
Total	14,220,000	14,220,000	2,279,387
Non-World Bank Financing			
Total	0	0	0
Total Project Cost	14,220,000	14,220,000	2,279,387

KEY DATES

Approval	Effectiveness	MTR Review	Original Closing	Actual Closing
31-May-2017	21-Aug-2017	18-May-2021	30-Jun-2022	30-Jun-2022

RESTRUCTURING AND/OR ADDITIONAL FINANCING

Date(s)	Amount Disbursed (US\$M)	Key Revisions
04-Oct-2018	1.13	Other Change(s)
30-Mar-2020	1.30	Change in Loan Closing Date(s)

KEY RATINGS

Outcome	Bank Performance	M&E Quality
Unsatisfactory	Moderately Unsatisfactory	Modest



RATINGS OF PROJECT PERFORMANCE IN ISRs

No.	Date ISR Archived	DO Rating	IP Rating	Actual Disbursements (US\$M)
01	23-Oct-2017	Satisfactory	Satisfactory	0
02	07-May-2018	Satisfactory	Satisfactory	1.13
03	09-Nov-2018	Satisfactory	Moderately Satisfactory	1.13
04	10-May-2019	Moderately Satisfactory	Moderately Satisfactory	1.18
05	25-Nov-2019	Moderately Satisfactory	Moderately Satisfactory	1.18
06	01-Jun-2020	Moderately Unsatisfactory	Moderately Unsatisfactory	1.35
07	11-Jan-2021	Moderately Unsatisfactory	Moderately Unsatisfactory	1.35
08	18-Aug-2021	Unsatisfactory	Unsatisfactory	1.74
09	24-Mar-2022	Unsatisfactory	Unsatisfactory	2.04
10	29-Jun-2022	Unsatisfactory	Unsatisfactory	2.13

SECTORS AND THEMES

Sectors

Major Sector/Sector	(%)
Energy and Extractives	100
Energy Transmission and Distribution	50
Renewable Energy Solar	50

Themes

Major Theme/ Theme (Level 2)/ Theme (Level 3)	(%)
Human Development and Gender	100
Gender	100
Urban and Rural Development	100
Rural Development	100
Rural Infrastructure and service delivery	100



Environment and Natural Resource Management	100
Climate change	100
Mitigation	100
Energy	100
Access to Energy	100

ADM STAFF

Role	At Approval	At ICR
Regional Vice President:	Victoria Kwakwa	Manuela V. Ferro
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I. PROJECT CONTEXT AND DEVELOPMENT OBJECTIVES

A. CONTEXT AT APPRAISAL

Context

1. **The Republic of Vanuatu (“Vanuatu”) is an archipelago of 82 volcanic islands (65 of them inhabited) that stretch for 1,300 km and cover a total land area of about 12,200 km².** They lie about 800 km west of Fiji and nearly 1,800 km east of Australia. At the time of appraisal, the total population of Vanuatu was approximately 270,000, comprising approximately 55,000 households, and was growing at roughly 2.5 percent per year¹. The majority of the population (roughly 75 percent) lived in rural areas. The Republic of Vanuatu was founded in 1980 from the New Hebrides, upon independence from the British and French administrations.

2. **Vanuatu was classified as a Small Island Developing State (SIDS), with a GDP per capita estimated at US\$2,805 in 2016.** After a previous decade of strong economic growth, the economy was stabilizing after the devastating Cyclone Pam² in March 2015 and a severe drought during 2015 and 2016. The economy was dominated by services including tourism (67.7 percent of GDP), agriculture (20.6 percent) and industry (11.7 percent). Exports included copra and coconut oil, canned tuna, cocoa, kava and timber.

3. **Grid-based electric power had reached roughly 30 percent of the population.** Four islands had grid systems (covering part of the island in each case) - Port Vila on Efate, Luganville on Santo, and on small parts of the islands of Malekula and Tanna. Two private utility concessionaires, Union Electrique de Vanuatu (UNELCO) and Vanuatu Utilities and Infrastructure Ltd (VUI), undertook generation, distribution, dispatch, billing and settlement for customers in their respective areas.³ The total installed generating capacity in these systems was 31.5 MW, of which 26.5 MW was in the Port Vila concession. Electricity generation on the grid systems was primarily from diesel (77 percent of generation), with smaller contributions from hydropower (10.7 percent), wind power (7.5 percent), coconut oil powered diesel generator units (4.4 percent) and solar PV (0.2 percent).⁴ There were approximately 22,300 households (40 percent of the population) in the utility concession areas at the end of 2015, although not all of these

¹ The population of Vanuatu is estimated to have grown to 315,700 in 2022, which would correspond to roughly 63,000 households.

² Cyclone Pam was a Category 5 storm that caused a reported 24 deaths and an estimated economic loss and damage of US\$450 million equivalent to 64 percent of Vanuatu’s GDP

³ UNELCO (60 percent owned by the French utility ENGIE) had been operating in Vanuatu’s electricity sector since 1939 and was the sole concessionaire up to 2010. GoV selected VUI to operate the expiring Luganville concession in 2010, however the process was contested by UNELCO and the Vanuatu Supreme Court in 2014 declared the VUI concession to be void due to selection process non-conformance with the Government Contracts and Tenders Act. GoV subsequently signed a concession contract with VUI in 2019 for the supply of electricity to certain parts of the islands of Santo, Maewo, Vanua Lava and Ambae. UNELCO’s concessions in Tanna and Malekula ended in mid-2020, and the Vanuatu National Provident Fund (VNPF), a compulsory national pension scheme, emerged as the new concessionaire in 2022. (UNELCO continues to be the power generator (with a shareholding change to 70 percent Vanuatu state ownership), and a local company Vate Electrics became the distributor).

⁴ Vanuatu has good solar irradiation. Estimates of average global horizontal irradiation (GHI) in various locations range from approximately 1,900 to 2,300 kWh/sq.m/year.



were yet connected to the grid, partly due to the relatively high cost of connection payable by the customers.

4. **Once connected to the grid, the domestic demand for electricity was strikingly low**, with an average monthly household usage of between 30 kWh (Malekula) and 54 kWh (Port Vila). This is due in part to the highly tiered tariff structure. For the UNELCO concessions, the first 60 kWh per month was charged at 0.34 times the base price⁵, the next 60 kWh per month at 1.21 times the base price, and any usage above that level at 3 times the base price. The grid-based service was therefore relatively affordable, with low-income residential customers spending between 2.3 and 3 percent of their incomes on electricity.⁶

5. **The Vanuatu Utilities Regulatory Authority (URA), established in 2008, was (and remains) the regulator of electricity and water network provision in Vanuatu.** URA approved the tariffs for Luganville electricity concession awarded to VUI in 2010, with a tiered tariff structure similar to that of UNELCO. URA assisted in resolving consumer complaints and advised the government on policy and legislative matters related to electricity and water.

6. **The Vanuatu National Energy Road Map (NERM) articulated Government of Vanuatu (GoV) energy policies and investment strategies up to 2030** with a particular focus on improving energy access. It was approved by the Council of Ministers in June 2013 and revised in June 2016. The NERM identified five priority pillars for the energy sector: access, petroleum supply, affordability, energy security, and climate change. It included an agenda of 41 key actions that were labeled either (i) “immediate”, including the Vanuatu Rural Electrification Project, Stage I (VREP I), which was already under implementation in mid-2016, or (ii) “highest priority”, including the Vanuatu Rural Electrification Project, Stage II (VREP II).

7. **Household electricity access in off-grid areas was estimated by GoV at only 9 percent in 2015.**⁷ Solar lanterns had been provided to rural households under previous donor-supported projects, however GoV and NERM did not consider that solar lanterns provided sufficient access to contribute to the NERM’s electricity access targets (revised to 60 percent for 2020 and 100 percent for 2030. However small “pico-SHS”, even from 5 watts-peak (Wp) upwards, could provide a valuable and improved service in rural Vanuatu.⁸ The main rural domestic customer demand was for lighting and phone charging. Some rural domestic consumers were deemed to be willing and able to pay for services such as television (TV) and refrigeration.⁹ However, affordability was the biggest barrier for consumers to move up to SHS¹⁰ with

⁵ The “base price” for the UNELCO concessions was equivalent to roughly 55 US cents/kWh during 2016.

⁶ SREP Investment Plan (November 2014), Table 2.1, page 14.

⁷ Per the NERM (June 2016 edition), Section 3.2. This was before VREP I started to make substantial progress in improving access to rural off-grid households. Reported sales of VREP I SHS were 1,071 in 2016; 3,396 in 2017; 7,850 in 2018; and 11,440 in 2019, totaling 23,757 units of which 21,207 were verified by DoE (per Annex 8 of the VREP I ICR). Given VREP I closed on September 30, 2020, there would have been additional sales during 2020. This was well above the initial target for VREP I of providing access to 17,500 households (that was revised up to 20,500 households). VREP I provided subsidized electricity access to at least 70 percent of Vanuatu’s off-grid households, albeit at a basic level for the most part and with a product warranty period of only two years.

⁸ NERM (June 2016), page 16.

⁹ SREP Investment Plan (IP) November 2014, Section 2.5.2. It was however noted in the IP (footnote 41) that some grid-connected households in some parts of the Luganville concession area had not increased their very low electricity usage even 20 years after the grid service first became available to them.

¹⁰ SHS (in this context) are a system consisting of photovoltaic (PV) panels, charge controllers, batteries and may also include an inverter. In the case of VREP I, around 35 percent of the systems sold included TVs (from 19-inch up to 49-inch).



higher power outputs and enhanced user benefits. The average total rural household income was Vanuatu Vatu (VUV) 79,500 (US\$750) per month, of which an average of VUV 69,300 (US\$630) was spent on total expenditures per household per month. On average, only 59 percent of a rural consumer's income was in the form of cash. (PAD, para 13).

8. **VREP I¹¹ was approved by the World Bank (WB) on November 7, 2014 with the project development objective to scale up access to electricity services for rural households, aid posts and community halls located in dispersed off-grid areas.** Specific targets were to facilitate the sales of SHS to 17,500 dispersed rural households (approximately 85,750 people), 230 aid posts, and 2,000 not-for-profit community halls located in dispersed off-grid areas. This would be achieved by subsidizing the retail cost of such systems through participating registered vendors¹². The project would focus on PV systems between 5 Wp and 100 Wp capacity that are 'plug and play' type, providing SE4All Multi-Tier Framework Tier 1 level of electricity services.¹³ These systems provide multipoint lighting, telephone charging, radio services and sometimes a TV.¹⁴

9. Other WB-supported projects.¹⁵ The Global Partnership for Output-Based Aid (GPOBA) Improved Electricity Access Project – to subsidize 4,375 new household connections across the grid areas of Vanuatu in partnership with the two utility concessionaires – was under implementation and closed on December 31, 2018.

Theory of Change (Results Chain)

10. The activities of the project were designed to provide a suitable means of electricity access incorporating primarily renewable energy (RE) to rural off-grid households as well as public institutions and businesses in Vanuatu. Component 1 would provide solar home systems (SHS) and micro grids to households and public institutions through a subsidized vendor model. Component 2 would provide electricity access to five rural villages through grant-funded mini grids that would then be operated on a commercial basis. Component 3 would provide important technical assistance to help ensure the smooth

¹¹ WB project number P150908. A US\$4.7 million grant was provided by the New Zealand Ministry of Foreign Affairs and Trade (NZMFAT) through the Pacific Regional Infrastructure Facility (PRIF). US\$3.1 million of the grant resources were earmarked for subsidy contributions and the other US\$1.6 million for TA and project management. VREP I was closed on September 30, 2020.

¹² Registered vendors (in this context) are sellers of products/systems within the territory of Vanuatu that are registered as authorized sellers and have entered into a Subsidy Implementation Agreement with the Recipient. Vendors were required to have previous experience in supplying SHS or micro grids within Vanuatu or in countries similar to Vanuatu having communities on remote islands or remote regions. Manufacture of product/system components within Vanuatu was not required. Four vendors were successfully registered under VREP I – see next footnote below. (A fifth vendor was registered and then dropped under controversial circumstances that contributed to a worsening of the relationship between GoV and the WB).

¹³ SE4All = Sustainable Energy for All. ESMAP (Energy Sector Management Assistance Program). 2015. *Beyond Connections. Energy Access Redefined*. Technical Report 008/15.

¹⁴ "Plug-and-play" solar systems are portable systems that are purchased as a kit and do not require specialist technical knowledge or personnel to install them. They can provide lighting, phone and small appliance charging and power a DC fan and a television. The size of "plug-and-play" systems was initially limited to between 5 Wp and 30 Wp to target the very remote communities. The size of products would be increased over time while remaining suitable for installation and maintenance by the users. The SHS offerings of the four approved vendors were sized between 6-38 Wp (PCS), 6-55 Wp (E-Tech), 29-200 Wp (Computer World) and 20-50 Wp (Savvy Solar). Across all the four vendors, 14,734 sales (62.2 percent of all sales) were sized 30 Wp or lower and around 26 percent of all sales were in the smallest size of 6 Wp. 906 sales (3.8 percent of all sales) were sized 120 Wp or higher.

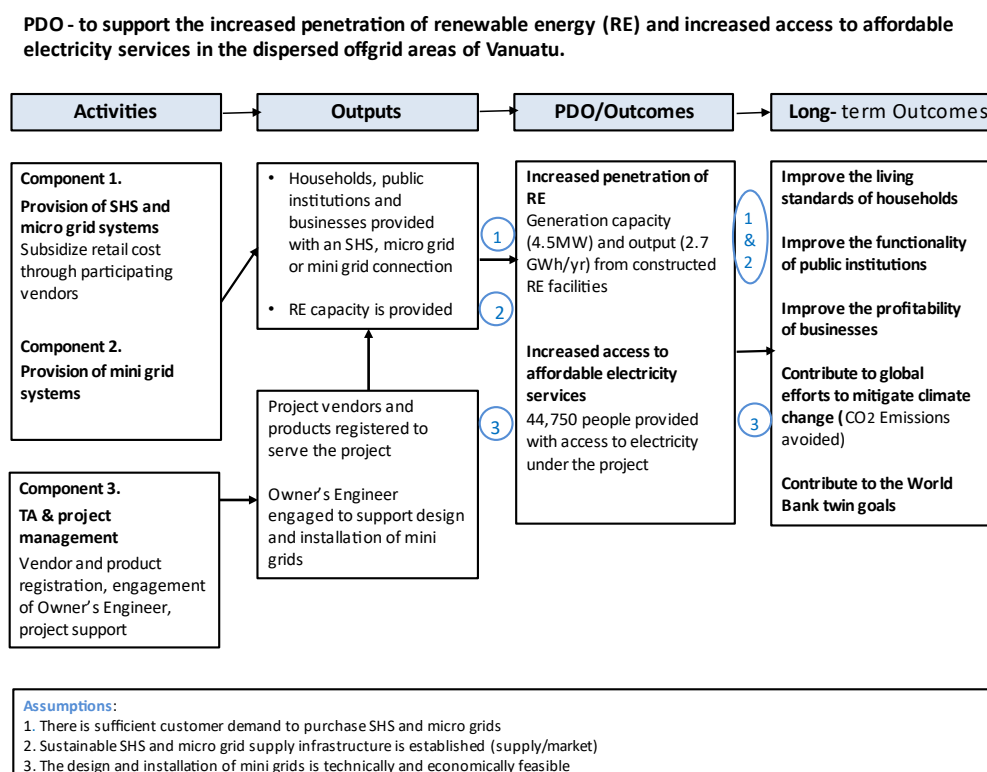
¹⁵ There were many donor-supported projects in the Vanuatu energy sector. See Table 3.11 of the SREP Investment Plan.



operation of the subsidized vendor model and provide technical supervision of the mini grids design and installation through the engagement of an owner's engineer. Once installed and operational, the SHS, micro-grids and mini grids would provide increased penetration of RE and increased access to affordable electricity services. The increased access to affordable electricity services to the population, public institutions and businesses would lead to the improvement of the living standards of the population, the improved functionality of public institutions and the improved profitability of businesses. The increased penetration of RE would contribute to global efforts to mitigate climate change.

11. The key assumptions are: (1) there is sufficient customer demand to purchase SHS and micro grids, which is particularly relevant since VREP I was using a similar subsidized vendor model to provide SHS to the rural population and there was uncertainty around the additional demand and uptake for VREP II; (2) sustainable SHS and supply infrastructure is established (supply/market), which would include suitable pre-financing offerings to improve end-customer affordability and allow the benefits of the project to be scaled up in a commercial manner after the subsidy scheme ended and thereby achieve the long-term objectives; and (3) the design of the mini grids is technically and economically feasible, which is important given the questionable track record of prior donor-supported attempts to introduce mini grids into Vanuatu.

Figure 1. Theory of Change of VREP II





Project Development Objectives (PDOs)

12. According to the VREP II Financing Agreement, the objective of the Project is to “support increased penetration of renewable energy and increased access to affordable electricity services in the dispersed off-grid areas of Vanuatu”. The definition in the VREP II Project Appraisal Document (PAD 2044) was the same except that it omitted the word “affordable”. The PDO can logically be separated into two distinct (although closely linked) objectives:

- PDO 1: Support the increased penetration of RE in the dispersed off-grid areas of Vanuatu
- PDO 2: Support the increased access to affordable electricity services in the dispersed off-grid areas of Vanuatu

Key Expected Outcomes and Outcome Indicators

13. Progress was measured against the following three PDO-level results indicators:

- PDO 1.1: Generation capacity of constructed RE facilities (MW). [Target: 4.5]
- PDO 1.2: Annual electricity output from RE as a result of RE constructed under the project (GWh). [Target: 2.7]
- PDO 2.1: People provided with a new electricity service (number). [Target: 44,750]

14. The progress toward the achievement of the PDOs was to be monitored through the following nine intermediate results (IR) indicators:

- IR 1 - Households received a SHS or micro grid (number) [Target: 8,400]
- IR 2 - Public Institutions received a SHS or micro grid (number) [Target: 37]
- IR 3 - Number of mini grids constructed under the project [Target: 5]
- IR 4 - Number of People connected to electricity through a mini grid constructed under the project [Target: 2,750]
- IR 5 - Direct project beneficiaries, percent of which are female [Target: 49]
- IR 6 - Total households of which are female headed households received a SHS, micro grid or electricity through a mini grid (percent) [Target: 13]
- IR 7 - Participants in consultation activities during project implementation (number) [Target: 2,000]
- IR 8 - Participants in consultation activities during project implementation who are female (number) [Target: 980]
- IR 9 - Grievances registered related to delivery of project benefits addressed (percent) [Target: 100].

15. The PDO and IR indicators were not changed during project implementation.

Components



16. **Component 1: Provision of SHS and micro grid systems in rural areas of Vanuatu (Approved US\$5.37 million, Actual US\$0.3 million).** This component targeted approximately 8,400 rural households (equivalent to 42,000 people) and 37 public institutions through a registered vendor model that was already being used on VREP I. Subsidy-eligible SHS for VREP II were approved by DoE in four product size ranges: 250 Wp (± 150 Wp), 610 Wp (± 200 Wp), 1,050 Wp (± 250 Wp) and 1,600 Wp (± 300 Wp).¹⁶ They were offered with DC output only, AC output only, or both. Subsidy-eligible micro grid systems were approved in four size categories: 2,200 Wp (± 300 Wp), 3,000 Wp (± 500 Wp), 4,100 Wp (± 600 Wp) and 5,600 W¹⁷(± 900 Wp).¹⁸

17. **Component 2: Construction of mini grid¹⁹ systems in rural areas of Vanuatu (Approved US\$6.8 million, Actual US\$0 million).** This component, anticipated to comprise 5 mini grids in the initial (pilot) stage, would support electricity service provision at an anticipated SE4All Tier 4 level to approximately 550 rural village households, which equates to around 2,750 people, and public institutions and businesses. The envisaged mini grids typically involve PV arrays, a battery bank able to store one day of expected demand and/or a diesel genset to meet peak demand and restart the system (that could be fueled by biodiesel from locally produced crops such as copra), and an inverter to supply alternating current (AC) to the village distribution network. Financial viability of the mini grids was identified as an issue during project appraisal, so the mini grids would need to be subsidized while balancing the impacts on the Government, the service providers, the end users (beneficiaries) and the communities. Rollout of mini grids beyond the initial stage would be subject to recalibration and lessons learnt from the piloting phase.

18. **Component 3: Technical Assistance (TA) and Project Management (Approved US\$2.0 million, Actual US\$1.7 million).** This component addressed: (i) establishing the vendor registration model for Component 1, (ii) engagement of an Owner's Engineer for Component 2 and (iii) project management.²⁰

¹⁶ Subsidy Implementation Manual (SIM), Rev 9, Section 8.4. The actual sizes within each product size category could vary within wide ranges, (e.g.) an individual SHS in the smallest 250 Wp size category range could theoretically be sized anywhere between 100 Wp and 400 Wp. Each vendor was allowed to offer three different configurations from each size category in the Product Catalogue published by DoE. In the April 2020 version of the Product Catalogue, only two product sizes were shown for sale in this category – 150 Wp and 275 Wp. A 150 Wp DC system without an inverter was offered at a subsidized price of VUV 56,000 excl VAT (equivalent to US\$465). A 150 Wp system with an inverter was offered at a subsidized price of VUV 70,000 excl VAT (equivalent to US\$580). No payment plans were included. SHS with up to two “standard” solar PV modules (sized up to 390 Wp) could be sold as pre-assembled kits and then installed by the purchasers if specific conditions were met as specified in the SIM (Section 8.5). Otherwise, SHS vendors were required to use a qualified installer, and the installation price could qualify for the same level of subsidy (33 percent in the case of VREP II Component 1).

¹⁸ SIM, Rev 9, Section 8.3.

¹⁹ In the Pacific context, a “mini grid” can be defined as a power system architecture that includes electricity generators, energy storage systems interconnected with a distribution system that supplies the entire electricity demand of a localized group of customers (typically a village) on an islanded basis. A “micro grid” provides similar architecture but at a much smaller scale and for only a single institutional campus or a few contiguous customers (e.g. school, staff quarters, local shop or health center).

²⁰ In addition, there was an allowance for Government contribution (US\$1.5 million “in kind”) to cover the GoV's direct project related costs, such as arrangements for voluntary land donation and other inter-departmental support.



19. **Table 1** below presents the indicative project financing plan provided in the PAD.

Table 1: Project Component Costs by Financing Sources
(US\$ million equivalent)

Project Component	Cost	IBRD or IDA Financing	Trust Funds
1: Provision of SHS and Micro grids in rural areas	5.37	1.5	3.87
2: Construction of Mini grids in rural areas	6.8	2.5	4.3
3: Technical Assistance and Project management	2.0		2.0
Total Project Costs	14.17	4.0	10.17

B. SIGNIFICANT CHANGES DURING IMPLEMENTATION

Revised PDOs and Outcome Targets

20. None.

Revised PDO Indicators

21. None.

Revised Components

22. None.

Other Changes

23. The Project was restructured twice, as indicated below.

24. The first restructuring agreement, dated October 4, 2018, linked the Sustainable Energy Financing Project (SEFP)²¹ to VREP II. However, the SEFP was not implemented in Vanuatu, therefore these restructurings had no effect.

²¹ WB project number P098423. The SEFP was a regional project that had concentrated on Fiji since 2014. It incorporated a Risk Sharing Facility (RSF) that provided partial loan guarantees by private and public financial institutions to support access to finance for consumers (households and businesses) for their cost contribution and to finance working capital for suppliers. However, the guarantee support from SEFP was not actually used for VREP I or VREP II. There were delays in executing the Implementation Agreements between Fiji and Vanuatu, issues associated with new terms of conditions of the 2018 restructuring and difficulties in executing the legal agreements with the Participating



25. The second restructuring, dated March 30, 2020, extended the closing date of the Pacific Regional Infrastructure Facility (Trust Fund) PRIF TF for two years to match the Project's revised closing date on June 30, 2022. It was logical to do this to avoid premature termination of this grant funding tranche.

26. An additional Project restructuring was discussed during late 2021 and early 2022 to change the scope of Component 1 and extend the Project for two years. It had been agreed to terminate the vendor delivery model for Component 1 by the end of 2021, based on the discussion during the Mid-Term Review (MTR) that took place in May-July 2021, with a view to transitioning this to a grant-based bulk procurement model to electrify public institutions. However, this restructuring was not completed due to unresolved differences between GoV and the WB, culminating in a request by GoV in April 2022 to use Vanuatu's national procurement procedures instead of those of the WB, that could not be accommodated by the WB. The Project closed as originally scheduled on June 30, 2022.

Rationale for Changes and Their Implication on the Original Theory of Change

27. PRIF TF was extended until June 30, 2022 in order that the project could continue to avail of these valuable grant resources. The changes did not impact the original theory of change.

II. OUTCOME

A. RELEVANCE TO PDOs

28. The relevance of the PDOs is assessed as **Substantial**.

29. The World Bank Group (WBG) Regional Partnership Framework (RPF) (FY17–FY21)²² was in advanced draft form at project approval and was published five months later in October 2017. It was stated in the RPF that the majority (70 percent) of the households in Vanuatu outside the two main cities did not have electricity, and that this level of consumption-based poverty was an indicator of deprivation, the remedy of which is a key objective of the WBG and the RPF. Explicit reference was made to VREP in the RPF. Increased access to basic services and improved connective infrastructure was one of its objectives (Objective 4.2). The RPF described disparities in basic services, including electricity, which were particularly pronounced among the Pacific Island Countries, and stipulated that closing the infrastructure coverage gaps was a key priority, including in Vanuatu. The project design was intended to address these goals by providing rural households and businesses and public institutions with electricity access through individual SHS, micro grids and mini grids. While the RPF did not specifically focus on RE, as a practical matter, the expansion of off-grid electricity access in Vanuatu is basically synonymous with expanding RE (primarily solar).

30. The RPF (FY17-FY21) had not been replaced or updated as at the Project closing date in June 2022, and is therefore assumed to continue to be relevant to the Project objectives. Given the enormous

Financial Institutions (PFIs), and subsequently the general economic downturn and increase in lending risk arising from COVID-19. Following the onset of Covid-19 there was limited interest from the previously participating banks - the Australia and New Zealand Banking Group Limited (ANZ), the Fiji Development Bank (FDB) and the Home Finance Corporation (HFC) - to recommence lending under the SEFP.

²² Covering nine countries in the Pacific including Kiribati, Nauru, the Marshall Islands, the Federated States of Micronesia, Palau, Samoa, Tonga, Tuvalu, and Vanuatu.



benefits of providing affordable electricity access to the rural population, including improving the standard of living of the population and enhancing the rural economy, the relevance at closing should not be in any doubt.

31. Both of the Project's objectives were in full alignment with Vanuatu's priority energy policies, based on the premise that access to modern energy is a fundamental enabler and catalyst for economic development, and enhancing the livelihoods and wellbeing of all Vanuatu citizens. The Project would contribute to increased access and affordability of electricity for communities in rural Vanuatu, in line with the key targets for electricity access and RE as identified in the updated NERM targets for 2030: (i) modern electricity access to 100 percent of households and public institutions, and (ii) all electricity to be generated from renewable sources. The updated NERM (June 2016) contained five green growth objectives. Objective 1 is to promote green energy as a catalyst for sustainable development. Objective 3 is to promote the use of RE in Vanuatu's main economic sectors. The Project objectives are directly linked and relevant for the above-mentioned green growth objectives.

B. ACHIEVEMENT OF PDOs (EFFICACY)

Assessment of Achievement of Each Objective/Outcome

32. The key indicators considered in this ICR to assess the achievement of the objectives are the three PDO-level indicators and two intermediate indicators that were used throughout the implementation of the Project. These five indicators are detailed in the text below and in Table 2 below.

PDO 1: Support the increased penetration of RE in the dispersed off-grid areas of Vanuatu.

33. Achievements per the relevant key indicators were:

- [PDO 1.1]: 0.29 MW of new RE capacity (6.4 percent of the target of 4.5 MW);
- [PDO 1.2]: 0.32 GWh of annual RE generation (11.8 percent of the target of 2.7 GWh).

34. The increased penetration of RE due to the Project was through the installation and operation of SHS and RE-powered micro grids and mini grids. Only 370 SHS were installed and operated under the Project as compared to the target of 8,400 per IR.1. No mini grids were installed or operated under the Project.

PDO 2: Support the increased access to affordable electricity services in the dispersed off-grid areas of Vanuatu.

35. Achievement per the relevant key indicators were:

- [PDO 2.1]: 3,625 individuals were provided with an electricity service (8.1 percent of the target of 44,750 individuals).
- IR1: 370 households received a SHS or micro grid (4.4 percent of target of 8,400)
- IR3: No mini grids were constructed under the Project (zero percent of target of 5)

36. The increased access to affordable electricity services due to the Project was to be achieved through the installation and operation of SHS and micro grids and mini grids. Only 370 SHS were installed



and operated under the Project as compared to the target of 8,400 per IR.1. No mini grids were installed or operated under the Project.

37. The achievement of the targets is summarized in Table 2 below. Penetration of RE could in theory be assessed by measuring the share of RE in the total power supply, however in this case the off-grid RE from SHS installations is not comparable in quality of service to on-grid RE, and also the project was not able to control the addition of non-project RE additions that would be needed to measure the share. Therefore, the indicators that were used are considered to be appropriate to assess the increased penetration of RE. The key indicators are considered to be fully relevant and attributable to the project.

38. Although the Project did strengthen three local energy retailers to incorporate quality standards on solar products, encourage product warranties and establish customer grievance and subsidy mechanisms, there was minimal achievement of the PDOs. The efficacy as measured by the extent of achievement of each of the PDO-level objectives is rated as **Negligible**.

Table 2: Final Targets and Achievement of Outcomes

Detail on plausible causal relationships between activities and outcomes

Objective/Indicator	Baseline	Target	Result at Closing	Achievement assessment
1. Support the increased penetration of RE in the dispersed off-grid areas of Vanuatu				
1.1 Generation Capacity of Constructed RE Facilities	0	4.5 MW	0.29 MW	Negligible (< 10 percent)
1.2 Annual Electric Output from RE as a Result of RE Constructed by the Project	0	2.7 GWh	0.32 GWh	Minimal (10-15 percent)
2. Support the increased access to affordable electricity services in the dispersed off-grid areas of Vanuatu				
2.1 People Provided with a New Electricity Service	0	44,750	3,625	Negligible
IRI 1. Households Received a SHS or Mini Grid	0	8,400	370	Negligible
IRI 3. Number of Mini Grids Constructed Under the Project	0	5	0	None

39. Almost all of the PDO-level targets (including 42,000 out of the 44,750 individuals to be provided with electricity) were expected to be attained through a vendor model to electrify individual households and public institutions. It was virtually identical to the VREP I vendor model except for the sale of larger SHS and micro grids. A small portion of the targets was to be achieved through a grant-funded model for the construction of up to five village mini grids and their subsequent tariff-supported operation.

40. Registered vendors sold 370 SHS to households against the target of 8,400 (i.e. 4.4 percent of the IR.1 target) and 104 SHS and micro grids to public institutions against the target of 37 (i.e. 280 percent of the IR.2 target).



41. The Project would partly subsidize the retail cost of SHS and micro grids by reducing the retail price paid by the customers and subsequently reimbursing the subsidy amount to the vendors for verified sales. A subsidy level was established (initially 33 percent of the gross retail price exclusive of VAT) with the intention that it could be varied over time depending on the take-up rate and the size of the systems. The consumer would pay for the system either by means of upfront cash or through a payment plan that could include microfinance credit via a microfinance provider or through a “pay-as-you-go” (PAYG) scheme set up by the vendor. The gross retail price was not regulated, and the market relied on effective competition between multiple registered vendors offering products and pricing that were transparently specified in a Product Catalogue maintained by DOE.²³

42. Changes to the wording in the Subsidy Implementation Manual (SIM)²⁴ had already been made for VREP I in March 2016 where it had defined the Beneficiaries as specifically for those “outside of the four electricity concessions” to now be those “outside and within the four concessions that are not likely to have a grid connection within the next four (4) years”. This amendment was requested by DoE after receiving feedback from the vendors of the increasing demand from potential customers within the concession areas that were not being prioritized for utility connection; having undertaken field visits to these areas and received feedback and noting the difficult natural terrain and its unprofitable challenge to extend grid connection; and having considered UNELCO and VUI’s planned grid extension in the next four years of VREP I. In addition, DoE was mindful of the ambitious SHS deployment target set to be achieved for VREP I by December 2019 and believed that the risk of not achieving the target could be reduced with a change in the restriction to the Beneficiaries to include those within and outside of the concession areas where grid connection was unlikely in the next four years.

43. The subsidized SHS product suite for VREP I had extended to systems as large as 200 Wp and VREP I had over-achieved its (already upwardly revised) sales target for households by its closing date of September 2020. The initial level of SHS subsidy for VREP I (50 percent of the gross retail price excluding VAT) was determined through an assessment of Willingness to Pay, average cash income levels for rural households and the expected retail costs of systems. The subsidy level was not revised during the VREP I implementation period. The VREP I sales accelerated markedly after the registration of an additional vendor that offered attractive payment plans. However, this vendor did not apply for registration under VREP II. Only three vendors were registered under VREP II. These vendors were working with a lower subsidy percentage than for VREP I (33 vs. 50 percent) and, more importantly, did not provide attractive payment plans to improve the affordability of the more expensive products and systems offered under VREP II.

44. Based on the agreement reached between GoV and the WB following a very extended Mid-Term Review (MTR) process that began in February 2020 and concluded in July 2021²⁵, the vendor subsidy model for VREP II was discontinued as of December 31, 2021, due to the very low demand and the below-mentioned issues that had arisen on VREP I regarding ineligible subsidy expenditures that also threatened to impact VREP II, and no subsidized sales were allowed after that date. By the Project closing date of June

²³ It was anticipated that there would be at least 10 registered vendors for the SHS products sold under VREP I. (There was an IR indicator to this effect).

²⁴ The VREP I SIM was modified to include VREP II after VREP II was approved, hence it became a joint SIM for VREP I and VREP II and further modifications were later made to the joint document. The final version after VREP I closed was for VREP II only.

²⁵ Finalization of the MTR report was seriously delayed for several reasons, including outstanding VREP I issues such as ineligible uses of project funds, change of the WB Task Team Leaders, new information received related to component 2 of VREP II to be included in the MTR report, and ongoing discussions with the DoE on the above matters.



30, 2022, registered vendor sales of SHS products had reached only 370 households compared to the end target of 8,400 households.

45. A disagreement arose between GoV and WB over ineligible subsidy expenditures on VREP I due to (i) the subsidization of TVs that was deemed by the WB task team to be disallowed under the Project, and where the WB then indicated the need for remedial action in relation to “excess” subsidy amounts on TVs already paid, and (ii) the failure by DoE as Verification Agent²⁶ to adequately verify all the sales where subsidies had already been paid back to the vendors²⁷.

46. An agreement was reached between the WB task team and the DoE during the lengthy VREP II MTR to simplify project activities by discontinuing the vendor model by December 2021 and transition it to a bulk procurement of SHS and micro grids for public institutions in the health, education and public administration sectors under a restructured project. However, a request by GoV in April 2022 to use national procurement procedures for the mini-grids contract instead of WB procedures was not resolved and the project was not restructured or extended. Implementation of mini grids had been slowed by a two-year delay in hiring the owner’s engineer for numerous reasons including technical challenges in preparing the terms of reference. Soon after the owner’s engineer contract was signed, the COVID-19 pandemic induced travel restrictions decelerated its outputs significantly. Despite the delays, the initial outputs of the contract e.g., community consultation and site selection of the mini grids, concept design, economic and financial viability were completed. Malekula Island was selected for the proposed site for all the mini grids after it met the selection criteria and land availability confirmed with due consultation with the community. Economic viability of the five mini grids at Malekula was reviewed and found reasonably acceptable to the WB. The revised implementation plan for these five mini grids was to complete the bidding process and award the Service Provider contract by July 2022. The WB reviewed the bidding documents and provided its “no objection”. In April 2022, GoV communicated by letter to the WB that according to a decision made by the Council of Ministers, Vanuatu’s national procurement rules rather than the WB’s procurement rules would need to apply. The WB responded by letter at the end of May 2022 that this was not possible except for in-country procurements that did not solicit international bidders.

Justification of Overall Efficacy Rating

47. As achievement of the three PDO-level targets was between 6.4 percent and 11.8 percent, the overall efficacy rating is assessed as Negligible.

²⁶ The Independent Verification Agent (IVA) role, which was initially envisaged to be independent from the DoE, was approved by the WB to be carried out by a project-financed consultant hired by and working within the DoE who was concurrently performing DoE jobs. (VREP I ICR, para 23).

²⁷ In an Annex to its request for restructuring and time extension of the VREP II Project in November 2021, DoE set out and elaborated upon the following grievances relevant to VREP I and the overall program: (i) failures in governance from former WB Sydney Office executive and TTL in administering the VREP program; (ii) allegations of numerous dealings from the former TTL that advanced the interest of his family’s business under VREP; and (iii) incompetent Project design and unworkable/incompetently designed verification methodology. The subsidy ineligibility controversy was therefore one of a number of GoV grievances against some aspects of the WB involvement in the design and implementation of VREP I and VREP II. Due in part to the overlapping VREP I and VREP II implementation periods, some key issues impacting VREP I were not addressed in a manner mutually satisfactory to GoV and WB before they impacted VREP II also.



C. EFFICIENCY

48. The VREP II project provided access to 3,625 individuals and provided 290 kW of new RE capacity at a total project disbursement of US\$2.28 million, equivalent to US\$7.70 per Wp. For comparison purposes, the VREP I project provided access to approximately 100,000 individuals (21,207 customers) at a total project cost of US\$4,648,358. This comes to US\$219 per customer for an average system size of 27 Wp, or US\$8.00 per Wp. As indicated in VREP I's ICR, this was considered reasonable compared to successfully implemented rural electrification projects elsewhere, considering the additional challenges presented by the small and dispersed rural market across multiple islands and the remoteness of Vanuatu. (VREP I's ICR, paragraph 33). In terms of the project-supported cost per Wp actually installed, VREP II was very similar to VREP I. The access level achieved was much lower for VREP II than for VREP I, but the quality of service was likely to have been better due to the larger systems that were installed and operated.

49. An economic and financial analysis was set out in the PAD. A Levelized Cost of Energy (LCOE) analysis showed that SHS/micro grids were the least cost solution for most communities. This analysis was supplemented by an affordability assessment. Table 8 in the PAD (copied below) showed the calculated monthly costs of SHS in four size categories against the monthly cash incomes in each of the six provinces as well as a "rural" category. The analysis assumed that the capital costs were subsidized by 33 percent, and the householder contribution repaid/financed over 4 years at 10 percent. Using these assumptions, systems up to around 1 kWh/day (roughly 450 Wp DC) were calculated to be relatively affordable for most households (i.e. costing no more than 10 percent of household cash income) provided the end purchaser's capital outlay could be spread out over 4 years.

Table 3: Affordability Assessment (PAD, Table 8):
Monthly costs of SHS against monthly cash incomes only by province

Affordability SHS	Torba	Sanma	Penama	Malampa	Shefa	Tafea	Rural
Power Cube (0.5 kWh/day) as % of ann. Cash	7%	4%	7%	7%	2%	15%	5%
2 Panel Kit - 12 Volts (0.9 kWh/day) as % of ann. Cash	13%	8%	14%	7%	2%	15%	5%
4 Panel Kit - 12 Volts (1.6 kWh/day) as % of ann. Cash	24%	15%	26%	24%	8%	51%	18%
4 Panel Kit - 24 Volts (2.2 kWh/day) as % of ann. Cash	63%	40%	69%	65%	21%	137%	47%

50. It was clarified during the VREP I MTR process that a key factor in determining customer affordability and subsequent SHS sales was the availability of suitable payment plans, including "pay as you go" (PAYG) options. The VREP I SHS sales greatly accelerated after an additional vendor was registered that offered suitable payment plans. This vendor did not apply for registration for VREP II and the three registered vendors for VREP II did not offer suitable payment plans. Therefore, a key assumption underlying the affordability analysis for SHS for VREP II did not materialize during implementation, and the initial calculation methodology that explicitly includes optimistic payment plan assumptions cannot be used for updating the analysis at completion. The larger and more expensive VREP II SHS products without suitable payment plans were patently unaffordable for most of the prospective customers despite the 33 percent subsidy, and only 370 were sold and installed against a target of 8,400.



51. A useful comparison for the upfront affordability of grid connections was included in the GPOBA Improved Electricity Access in Vanuatu Project (P133701). It was estimated by an independent consultancy that consumers could afford between US\$110 and US\$170 for service connections, which represented from 31 to 48 percent of the average monthly income of low-income households. (ICR for the GPOBA project, February 2020, footnote 21). The cheapest subsidized SHS products under VREP II (150 Wp DC) were priced at approximately US\$465 plus 15 percent VAT, or roughly US\$550, so it is hardly surprising that so few were sold. Not only that, but the cheapest systems under VREP II were sold alongside similarly sized VREP I systems that were subsidized at a higher level (50 percent) and benefited from payment plans that were not available for the VREP II systems. The additional price associated with customized installation was likely to be a further disincentive for the sale of the larger VREP II systems (with multiple PV modules and batteries).

52. As explained above, it was not practical to use the same analysis methodology as used in the PAD to assess the affordability of the relevant products (SHS and micro grids). These were assumed in the PAD to benefit from an attractive payment plan that paid off the purchase price over 4 years at a 10 percent interest rate, however this benefit was not available in practice. Without the payment plan, the products were assessed to be unaffordable for most of the population.

53. For Component 2 (mini grids), the economic and financial analysis in the PAD led to the conclusion that the construction cost would need to be fully financed by the Project, and that a customer service tariff level at or near the base tariff for the existing grid service would be needed to support the ongoing operating and maintenance (O&M) costs of the mini grids. The preparation of the terms of reference for the owner's engineer for the mini grids was delayed as the scope would vary depending on the operating model (existing service providers or third-party operators), involving protracted discussions among DoE, URA and the service providers. Further delay was experienced in tendering for the owner's engineer. The owner's engineer recommended that one island (Malekula) was likely to be suitable for the mini grids. UNELCO was the service provider on Malekula, however its concession on the island expired without extension in July 2020, and it was in May 2022 that the new concessionaire was put in place for Malekula and Tanna islands. DoE was planning to award an O&M contract (Concession Deeds) to the new concessionaire, but the Council of Ministers decided not to proceed with a restructuring and extension using WB procurement procedures and the required project restructuring and extension did not happen. Therefore, no mini grids were procured or constructed, and an updated EIRR or FIRR is not calculable.

54. The prolonged efforts by both sides in resolving VREP I issues had a negative impact on the overall VREP program, that compromised the administrative efficiency of VREP II. The inclusion of ineligible items in the subsidy calculations also led to some of the subsidy expenses (approximately 11 percent of the total VREP I subsidies paid) being declared ineligible for project funding. Addressing the verification process and inclusion of ineligible items took a long time to resolve for VREP I and significantly diminished the administrative efficiency with which the VREP I and VREP II projects were implemented. This was a major reason for prematurely terminating the vendor program for VREP II before any significant sales volume could be achieved. This issue was also cited by GoV as a reason for requiring a switch to Vanuatu's national procurement procedures even though the VREP I ineligible expenditures were not directly related to procurement. Restrictions related to Covid-19 resulted in administrative delays and inefficiencies. Selection of sites for mini grids was delayed, the project manager was stranded in Australia for over a year and was unable to provide implementation support, and the WB team could not travel to Vanuatu on missions. Also, the Owner's Engineer charged additional cost for COVID delays. The WB did ensure that



additional human resources were provided in Vanuatu to review the subsidy payments for both VREP I and VREP II for fiduciary purposes or to support DoE on a day-to-day basis including by Port Vila-based staff.

Assessment of Efficiency and Rating

55. The Efficiency rating is assessed as **Modest**. It is assessed to be below what would be expected in the operation's sector. The deterioration of the working relationship between GoV and the WB on the overall VREP program (particularly related to the ineligible expenditure controversy) negatively impacted the Efficiency of VREP II, however its primary impact was on the Efficacy which has been rated Negligible. The WB made significant efforts to improve the relationship during the last 12-18 months of the VREP II implementation period. The proposed Modest rating for Efficiency is based on the comparable cost per unit of added SHS capacity to other off-grid electrification programs. The cost efficiency of the mini grids was not tested due to the termination of the project before any installation could be achieved. The factors that led to the poor outcomes for VREP II have largely been reflected in the assigned Negligible rating for Efficacy.

D. JUSTIFICATION OF OVERALL OUTCOME RATING

56. The Relevance was rated **Substantial**, the Efficacy was rated **Negligible**, and the Efficiency was rated **Modest**. The Overall Outcome Rating is therefore **Unsatisfactory**.

E. OTHER OUTCOMES AND IMPACTS (IF ANY)

Gender

57. The Project was designed as gender-neutral, however specific indicators were provided related to the gender of direct Project beneficiaries. A donor-supported organization in Vanuatu specifically sought to recruit women as entrepreneurs, including for the installation and maintenance of solar systems, and this had the potential for synergy with the VREP program.

Institutional Strengthening

58. The VREP program (VREP I and II) included substantial TA that was intended to support DoE in its mission of increasing electricity access in off-grid areas of Vanuatu. DoE uses the mechanisms established under the program to continue financing household access under the National Green Energy Fund established in 2018. At the time of writing, it remains to be seen whether much of this work (such as was done by the owner's engineer) will be beneficial to the DoE. **Mobilizing Private Sector Financing**

59. The vendor program for SHS featured private financing by the consumers (final sales) and the vendors (purchase of inventory). The TA included the promotion of vendor finance and other microfinance to accelerate sales by making the SHS products more affordable to customers, however this aspect of the TA was largely unsuccessful. One of the VREP II vendors used limited vendor finance to promote SHS sales, however the VREP II vendor finance was inferior to that offered under VREP I. This was likely due in part to vendors' capital and cash flow constraints having a greater impact on the larger and more expensive systems being sold under VREP II. An attempt was also made by WB to link a risk sharing (partial



loan guarantee) model used in Fiji to VREP I and VREP II. The project was not successful in mobilizing any significant private financing as evidenced by the very low SHS sales.

Poverty Reduction and Shared Prosperity

60. Given the negligible efficacy of the Project, it is unlikely that there was any material impact on poverty reduction and shared prosperity. However, over 100 public institutions did benefit from an enhanced electric service, a level that exceeded the relevant target.

Other Unintended Outcomes and Impacts

61. None.

III. KEY FACTORS THAT AFFECTED IMPLEMENTATION AND OUTCOME

A. KEY FACTORS DURING PREPARATION

62. The Project was part of a larger VREP program, that was approved in two stages. At the time VREP I was approved by the WB Board in November 2014, the WB in conjunction with GoV was finalizing a Scaling Up Renewable Energy Investment Plan (SREP IP) for consideration for funding through the Strategic Climate Fund. The projects under consideration as part of the SREP IP were (i) extension of the VREP I project to include micro grids and mini grids, and (ii) separately from the VREP, investment in two small hydro projects to increase RE in Vanuatu's electricity generation mix through the Asian Development Bank and providing access to villages through grid extension. The SREP IP was endorsed in November 2014. It referred to a first stage of the VREP using a vendor model which aligned with the actual VREP I project and a second concept stage to be funded by SREP which included larger systems for households and public institutions using a variety of potential delivery models including fee-for-service franchises, concessions and Renewable Energy Service Companies (RESCOs). The most appropriate models for specific local applications would be identified later as part of feasibility studies and transaction advisory engagements. However, there was not sufficient time to perform these studies and engagements prior to the WB Board approval of VREP II in May 2017.

63. The VREP I project design included substantial TA that was carried over to VREP II in many respects, including establishing vendor and product registration arrangements and support with the development of microfinance products. It also included funding for project management and related support, i.e. capacity building and support to DoE through technical experts and advisors, training for the DoE staff, awareness programs to rural communities, and verification of subsidy claims prior to payments. Because of the substantial TA program, the VREP I and VREP II project designs had very similar vendor registration programs and subsidy implementation and verification processes.

B. KEY FACTORS DURING IMPLEMENTATION

64. **VREP II proceeded into the implementation phase before important lessons could be learned from VREP I implementation and serious problems could be corrected.** The achievement of the VREP I household connections exceeding the project target (over 21,000 verified sales) suggests that this vendor and subsidy model worked well as far as the service delivery was concerned. Despite some significant positive outcomes for VREP I, there were several implementation challenges on VREP I that negatively affected VREP II. VREP I implementation was compromised by the identification of ineligible expenditures



that was a highly unfortunate outcome of this grant-funded project. Ineligible subsidy expenditures were identified on VREP I for two reasons:

- (i) the alleged improper subsidization of TVs that were packaged with over 5,000 of the SHS products, and
- (ii) the failure of the DoE to fully comply with the SHS sales verification methodology, resulting in a disallowance of some reported sales and related subsidy reimbursement payments that had already been made. While no TVs were included in approved VREP II products, the verification issue per (ii) above was the same for Component 1 of VREP II as for VREP I.

65. The SHS size for VREP I originally varied between 5 Wp and 30 Wp (per Project Paper and version 1 of the SIM), however systems up to 200 Wp became eligible for subsidy at a later stage of VREP I implementation. Only 906 (3.8 percent) of the products sold under VREP I were 120 Wp or larger, partly because the vendor selling these larger kits became embroiled in controversy over its subsidization of appliances that were packaged with the products. Indeed, the relevant subsidies were high for some of these products, reaching close to VUV 100,000 (US\$900) after adjustment. This vendor did not register for VREP II. The failure of larger products to sell well under VREP I would have portents for VREP II that was focusing on the sale of larger systems.

66. Even though it was contemplated in the design that the subsidy scheme could be varied as needed during implementation, this was not done in practice. DoE held a view that subsidy level could have been higher to address the affordability challenges. There was an expectation towards decreasing the subsidy level over time, which is an understandable “sustainability” narrative and is fine as long as the required starting levels are robustly assessed, and unit prices are falling over time. It was also evident that other local vendors were selling SHS products and systems without the VREP subsidies, which could put them at a disadvantage.

67. Supply chain and working capital issues for VREP I vendors were already evident by the time that VREP II began implementation in late 2017. The two vendors that were already registered for VREP I expressed difficulties with accessing working capital as well as long waits to receive products (some were required to pay in full in advance to then wait eight months to receive their products). Developing the supply chain and considering measures to support vendors with working capital and consumers through payment plans were suggested as ways to further enhance the project results. A further challenge would have been that the subsidy reimbursement to the vendors was due only after the subsidized sale to the customer was completed and verified, thus exacerbating the vendors’ working capital problem.

68. The VREP I MTR report provided further guidance on improving project management and communications, vendor and product qualifications processes, opportunities to “recruit” more vendors, opportunities for access to working capital and microfinance and if required, project re-design and possible extension to enable the achievement of the project objectives. Clearly most of these considerations were also highly relevant for the recently approved Component I of VREP II that used the same vendor model.

69. The proposed restructuring of Component 1 of the VREP II project that was requested by GoV in November 2021 involved the replacement of the vendor model by the supply and installation of solar power systems for more than 1,100 off-grid public sector buildings through a bulk procurement approach. This new approach presented its own complex challenges, particularly during the Covid-19 travel



restrictions that would exclude international companies from mobilizing for installation. As noted in the Efficiency section above, restrictions related to Covid-19 resulted in administrative delays and inefficiencies. Selection of sites for mini grids was delayed, the project manager was stranded in Australia for over a year and was unable to provide implementation support, and the WB team could not travel to Vanuatu on missions. Also, the Owner's Engineer charged additional cost for COVID delays. The WB did ensure that additional human resources were provided in Vanuatu to support DoE on day-to-day operations and to review the subsidy payments for both VREP I and VREP II for fiduciary purposes.

IV. BANK PERFORMANCE, COMPLIANCE ISSUES, AND RISK TO DEVELOPMENT OUTCOME

A. QUALITY OF MONITORING AND EVALUATION (M&E)

M & E Design

70. The RF and M&E design was somewhat based on emerging lessons from VREP I, for example the number of SHS vendors was dropped as a target. The baselines were clear and coherent, except in the case of the SHS sales volume where zero was probably not the correct number (given the significant contribution already underway from VREP I). Indeed, there would appear to be a strong rationale to use an Additional Financing (AF) of VREP I instead of approving and implementing a separate project for VREP II. The target for electrification of public buildings (health, education and public administration) looked unusually low at only 37 units, when there were well over 1,000 such units that were unelectrified. However, this was likely to be connected with the vendor model to be applied, that was less suited for mass electrification of public institutions, many of which would not have suitable budget and payment mechanisms to facilitate engagement with vendors. The M&E design was adequate to inform about the project progress and results.

M & E Implementation

71. As the implementing agency, the DoE was responsible for ensuring the timely and accurate collection of the requisite indicator data. The DoE collected and shared with the WB information as required in the RF to inform implementation progress. But they stopped reporting project implementation progress from early 2020 and therefore limited information is available on project implementation toward the closing of the Project. Based on the agreement during the MTR mission in May-July 2021, the vendor subsidy model was discontinued as of December 31, 2021. Much of DoE staff efforts, however, were absorbed by the remaining verification work and processing outstanding claims for both VREP I and II, that included a lengthy and complex process of recalculating the eligible levels of subsidy expenditures. The WB updated the results indicators on a regular basis as part of the six-monthly reporting cycle.

72. The very low demand for the VREP II SHS could have signaled the need to revise the relevant sales target and this was not done. However, the SHS demand for VREP I also started to build slowly and then markedly accelerated after the third year of implementation when an additional registered vendor came on board. The subsidy level and other parameters could also have been re-evaluated during implementation when the SHS demand was failing to materialize. The relative lack of vendor finance and other microfinance products to help achieve VREP II SHS product affordability was a key issue that should have received more attention, however it is not necessarily easy to deal with this issue in isolation due to



the diverse vendor capabilities and characteristics. The promotion of competition between multiple vendors will tend to lead to the enhancement of product offerings including pre-finance offerings.

M & E Utilization

73. In light of the failure of the Project to deliver significant outputs and results, the M&E utilization was rather a moot point.

Justification of Overall Rating of Quality M & E

74. The M&E quality is rated “Modest”. Though the M&E design was adequate by and large, there were significant shortcomings in implementation as indicated above.

B. ENVIRONMENTAL, SOCIAL AND FIDUCIARY COMPLIANCE

75. The Financial Management (FM) rating has been assessed as Moderately Unsatisfactory in the last two FM supervision missions conducted in year 2022. The overall FM arrangements in place for the Project met the minimum requirements as stipulated in Bank Policy - Investment Project Financing. The Project FM staff turnover during Project life impacted the compliance with the Project financial reporting requirements resulting in delayed submission of the audit reports, delayed preparation of the project financial statements, and IFRs being submitted after the deadline mostly due to lack of proper accounts reconciliation with the information in the Government Financial Management System – Smartstream. The Designated Accounts (DAs) showed a low disbursement rate during the Project life.

76. The Procurement rating was assessed as Unsatisfactory for the last 12 months of the project implementation and no contracts were signed after September 2020. At project close, contracts had been awarded amounting to only \$1.88M. Procurement plans were not updated on a regular basis, there were long delays in procurement decision making by DoE, and poor recordkeeping and contract management practices. A major restructuring and time extension of the Project to facilitate meeting its PDO did not proceed due to an unresolved disagreement between GoV and the WB regarding the procurement procedures to be used, particularly for the mini-grid procurements. DoE’s project management team was under-resourced and had budget overrun. The Project progress reporting was delayed. The Project implementation also suffered because the Project Manager could not provide on-site implementation support during a crucial time of the project implementation since he was stranded in his home country and could not return to Vanuatu due to COVID-19 induced border closing. Although the Project Manager’s contract ended in October 2021, there was delay in initiating a local replacement of the Project Manager position despite several discussions held between the DoE and WB as well as reminders.

77. **Safeguards Compliance.** Safeguards performance as of August 2021 was rated as Moderately Unsatisfactory; the PMU never recruited environmental and social (E&S) officers as per the legal agreement and as such, the client was unable to monitor and assist implementation of the E&S requirements. However, the Owner’s Engineer had relevant E&S capacity for supporting the design and siting processes for the mini grids.

78. The Project was rated as having Moderate environmental and social risk and was classified as a Category B (partial assessment) at appraisal, triggering the following World Bank operational policies:



- **OP 4.01 - Environmental Assessment** was triggered because some planned investments, particularly modular solar / battery / diesel hybrid systems, may generate minor to moderate site-specific and time-bound adverse environmental impacts. Activities were screened for environmental and social impacts and mitigation measures were put in place to address potential site-specific impacts such as noise, health and safety risks for workers and waste disposal.
- **OP 4.12 - Involuntary Resettlement** was triggered because project activities required land acquisition and/or leasing of sites selected by beneficiary communities.

79. A program-level ESMF was prepared and disclosed on December 15, 2016, before appraisal. The ESMF included guidance for candidate projects, provided a process for environmental impact assessments and the development of environmental management plans and included reference to the code of practice for disposal of batteries. A Resettlement Policy Framework was prepared during appraisal, however due to lack of E&S officers in the PMU, the Owner's Engineer took responsibility for implementing the requirements. A Grievance Redress Mechanism was established and no grievances were received during the project cycle.

80. The project activities (including SHS & mini grids and TA) were assessed to have environmental impacts that are site specific, temporary, and readily manageable through the project's ESMF application. The Environment and Social Impact Assessment (ESIA) for the construction of the mini grids has been completed. However, the construction of mini grids did not proceed. The adoption of regulations for disposal of solid wastes (batteries) did not progress despite substantial efforts by the DoE before the project closing date.

81. No occupational health and safety incidents have been reported during implementation.

C. BANK PERFORMANCE

Quality at Entry

82. The VREP II PAD mentions that the design builds on lessons from other projects. It was noted in the PAD that the VREP I vendor program (to be adapted for Component 1 of VREP II) was progressing satisfactorily at that time. The economic analysis in the VREP II PAD indicated the likelihood of issues with consumer affordability for SHS above 450 Wp or so at a 33 percent subsidy level, and this issue was not explicitly addressed in the design of the subsidy scheme. However, it was noted that 33 percent was an initial VREP II subsidy level that could be changed based on experience during implementation. There was a very low target for public buildings (37 buildings), and this segment was therefore relatively unserved by the initial project design. The fast approval of VREP II while VREP I was still at an early stage of implementation meant that key issues that emerged with the subsidy mechanism on VREP I could not be addressed upfront for VREP II.

83. As indicated in the VREP II PAD (paragraph 50), the proposed Implementing Agency (DoE) was implementing other WB-financed energy projects at that time, and was familiar with the WB policies and procedures for project implementation. Since 2011, the DoE staffing had grown to support implementation of multiple projects. The DoE was recruiting for a number of key positions which would further strengthen the current capacity of the department. In addition, the DoE staff was supported by



five consultants funded under WB-financed projects, including the Project Manager for VREP I & II. VREP II would be the largest project the Department has undertaken, and it was foreseen that the implementation of VREP II would stretch the capacity of DoE's staff and consultants. The Project would provide (under Component 3) project implementation support and TA to strengthen DoE's technical capacity, E&S safeguards oversight and its fiduciary capacity. The Project would support the recruitment of a technical specialist to support the vendor and product registration arrangements, communications and implementation arrangements and a verification agent to verify product sales to support implementation of Component 1 and an Owner's Engineer to support implementation of Component 2.

84. The overall risk was assessed as Substantial. In particular, the FM risk was identified as Substantial primarily due to the limited capacity of the DoE to handle subsidy grants. It was envisaged that additional FM capacity-building support would be provided under the project, but this was not done. The original sales verification procedure was overly demanding for any small agency such as the DoE due to the required volume of phone calls to customers, physical verifications and other required verification actions that were not fulfilled, with a result that there was inadequate verification of subsidized sales that led to a retroactive assessment of ineligible subsidy expenditures. (It was originally included in the Project design that an IVA would be engaged, which would likely have been more effective had it been retained).

Quality of Supervision

85. The WB conducted implementation support visits regularly at least twice per year. In many cases these were joint missions that covered VREP I and II. As was also observed with VREP I, the WB team was at times insufficiently staffed. Issues that emerged on VREP I, including trying to deal with the posited ineligible subsidy expenditures, consumed the attention of DoE and the WB project team, and the MTR process dragged on for over a year while dealing with pre-closing issues on VREP I in an environment where cooperation and trust between GoV and the WB had been seriously eroded. The WB appointed a co-TTL to increase team support, and in late 2020 a new WB task team was put into place. The new TTLs managed to complete the MTR that had been stalled and made a highly commendable effort to improve the relationship with DoE and rectify problems to the extent possible during the final 12-18 months of implementation, however it was not possible to restructure and extend the Project to make effective use of the grant and concessional resources available.

86. ISR Reporting was timely and detailed, however the usefulness was at times compromised by the inability of GoV and WB to reach agreed positions on some of the key issues impacting the implementation of the Project. The candor of the final ISR reflected the input from the new TTLs and task team that reached a more balanced assessment of the implementation and relationship issues facing the Project.

Justification of Overall Rating of Bank Performance

87. There were major shortcomings in quality of supervision, however mitigated by the fact that the new WB task team made a highly commendable effort to improve the relationship with DoE and rectify problems to the extent possible during the final 12-18 months of implementation. Overall, the Bank performance is assessed as "Moderately Unsatisfactory".

D. RISK TO DEVELOPMENT OUTCOME

88. In light of the negligible efficacy, there is little risk to development outcome.



89. A subsidy mechanism to support the uptake of solar products needs careful design incorporating market soundings and may need to be adjusted during implementation. A 50 percent retail subsidy had already been applied for VREP I with considerable success in facilitating vendor sales of “plug and play” solar products to rural off-grid customers. Simply providing a lower 33 percent retail subsidy for the larger VREP II SHS systems did not work well. This suggests that the analysis required is more complex and market-specific and that a flexible approach may be needed to adapt to the market dynamics and achieve specific objectives.

V. LESSONS AND RECOMMENDATIONS

90. **A subsidy mechanism to support the uptake of solar products needs careful design incorporating market soundings and may need to be adjusted during implementation.** A 50 percent retail subsidy had already been applied for VREP I with considerable success in facilitating vendor sales of “plug and play” solar products to rural off-grid customers. Simply providing a lower 33 percent retail subsidy for the larger VREP II SHS systems did not work well. This suggests that the analysis required is more complex and market-specific and that a flexible approach may be needed to adapt to the market dynamics and achieve specific objectives. The inclusion of appropriate pre-financing including vendor financing to improve end-customer affordability is a key success factor, and more success in this regard was achieved on VREP I that focused on smaller products and systems and achieved more competition between vendors that likely stimulated the provision of pre-financing.

91. **Given the rapid WB approval of VREP II after VREP I was approved and the sequencing of these very similar operations, an Additional Financing (AF) of VREP I could have been considered instead of a separate VREP II project.** An AF could have helped with consistency and reduction of administrative burden and could have allowed for the entire program to be extended and restructured with a more flexible and optimal deployment of the combined grant and concessional funding (assuming the ineligible subsidy expenditure issue could be resolved, as mentioned below). Also from the client perspective, there was essentially a single VREP program with a primary objective to utilize donor subsidy resources to improve electricity access in off-grid areas.

92. **In the case of a phased project or series of projects, key lessons learned from initial projects or project phases should be incorporated for the benefit of subsequent projects or phases.** Detailed registration and verification mechanisms were developed for the initial VREP I subsidy program, and these were used again for VREP II before lessons could be learned from VREP I. Significant problems were encountered in these areas on VREP I that also impacted negatively on VREP II. Where a subsidy program is being expanded to reach an additional population segment, lessons from the initial program are likely to be relevant, and sufficient time should be given to allow lessons to be learned from the initial program before implementing the follow-on program.

93. **Design of a verification system should be supported by sufficient human, time and financial resources to meet the verification workload, and the system design and resulting implementation workload should be proportionate to the client and country capacity while pragmatically upholding fiduciary requirements.** Given the intensive workload required by the verification system characteristics that were adopted in this Project, it would have been optimal to reduce the workload burden, such as through a detailed multi-stakeholder consultation and a process engineering evaluation that would



reduce the number of required verification actions to a manageable level, ensure that ineligible expenditures are minimized (including by appropriate adjustment of the criteria for subsidy expenditure eligibility and critical examination of identified causes of ineligibility), and maintain an overall sense of proportionality within appropriate fiduciary constraints given the small island context and the limited client capacity. A situation where the entire subsidy program was terminated due largely to ineligible expenditures arising from a contested verification process could and should have been avoided.

94. **The affordability of solar products to end-customers may rely more on the availability of suitable payment plans from vendors than on the extent of upfront subsidy provided, and this issue typically becomes more important as product/system sizes and prices are increased.** It was evident that the suite of payment plans offered by vendors for VREP II was less attractive than offered for VREP I, when the opposite should have been the case in order to improve the affordability of the larger and more expensive VREP II products for the rural population.

95. **Cash flow and capital constraints facing product vendors should also be taken into consideration for design and implementation of the Project.** Vendors in remote places such as Vanuatu may need to purchase and pay for systems and components from abroad several months before they are delivered to them for sale to end-customers in Vanuatu, creating issues with working capital. This in turn affects their ability to offer attractive payment terms to end-customers. Ongoing consultations with vendors and end-customers that includes discussion of challenges facing them is generally appropriate.



ANNEX 1. RESULTS FRAMEWORK AND KEY OUTPUTS

A. RESULTS INDICATORS

A.1 PDO Indicators

Objective/Outcome: Support the increased penetration of RE in the dispersed off-grid areas of Vanuatu.

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Generation capacity of energy constructed - renewable energy facilities	Megawatt	0.00 30-Jun-2017	4.50 30-Jun-2017		0.29 27-Jun-2022

Comments (achievements against targets):

- [PDO 1.1]: 0.29 MW of new RE capacity (6.4 percent of the target of 4.5 MW);

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Annual electricity output from renewable energy as a result of renewable energy constructed under the project	Gigawatt-hour (GWh)	0.00 30-Jun-2017	2.70 30-Jun-2017		0.32 27-Jun-2022



Comments (achievements against targets):

- [PDO 1.2]: 0.32 GWh of annual RE generation (11.8 percent of the target of 2.7 GWh).

Objective/Outcome: Support the increased access to affordable electricity services in the dispersed off-grid areas of V

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
People provided with a new electricity service	Number	0.00	44,750.00		3,645.00
		30-Jun-2017	30-Jun-2017		27-Jun-2022

Comments (achievements against targets):

- [PDO 2.1]: 3,625 individuals were provided with an electricity service (8.1 percent of the target of 44,750 individuals).
- IR1: 370 households received a SHS or micro grid (4.4 percent of target of 8,400)
- IR3: No mini grids were constructed under the Project (zero percent of target of 5)

A.2 Intermediate Results Indicators

Component: Provision of Solar Home Systems and Micro Grids systems in rural areas of Vanuatu

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Households received a Solar	Number	0.00	8,400.00		313.00



Home Systems or micro grids		30-Jun-2017	30-Jun-2017		27-Jun-2022
Comments (achievements against targets):					
Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Public Institutions received a Solar Home System or micro grid	Number	0.00 30-Jun-2017	37.00 30-Jun-2017		104.00 27-Jun-2022
Comments (achievements against targets):					
Component: Construction of Mini Grids systems in rural areas of Vanuatu					
Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Number of mini grids constructed under the project	Number	0.00 30-Jun-2017	5.00 30-Jun-2017		0.00 27-Jun-2022
Comments (achievements against targets):					
Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised	Actual Achieved at



				Target	Completion
Number of People connected to electricity through a mini grid constructed under the project	Number	0.00 30-Jun-2017	2,750.00 30-Jun-2017		0.00 27-Jun-2022

Comments (achievements against targets):

Component: Technical Assistance and Project Management

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Participants in consultation activities during project implementation (number)	Number	0.00 30-Jun-2017	2,000.00 30-Jun-2017		0.00 27-Jun-2022
Participants in consultation activities during project implementation - female	Number	0.00 30-Jun-2017	980.00 30-Jun-2017		0.00 27-Jun-2022

Comments (achievements against targets):

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Grievances registered related	Percentage	0.00	100.00		100.00



to delivery of project benefits addressed (%)		30-Jun-2017	30-Jun-2017		27-Jun-2022
Comments (achievements against targets):					
Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Total households of which are female headed households received a Solar Home System, micro grid or electricity through a mini-grid	Percentage	0.00 30-Jun-2017	13.00 30-Jun-2017		11.80 27-Jun-2022
Comments (achievements against targets):					
Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Direct project beneficiaries of which female	Percentage	0.00 30-Jun-2017	49.00 30-Jun-2017		50.00 27-Jun-2022
Comments (achievements against targets):					





B. KEY OUTPUTS BY COMPONENT

Objective/Outcome 1: Support the increased penetration of RE in the dispersed off-grid areas of Vanuatu	
Outcome Indicators	1. Generation capacity of constructed RE facilities 2. Annual electric output from RE as a result of RE constructed by the Project
Intermediate Results Indicators	1. Households received a Solar Home Systems or micro grids 2. Public Institutions received a Solar Home System or micro grid 3. Number of mini grids constructed under the project 4. Number of people connected to electricity through mini grids constructed under the Project (Note: these IR Indicators also support Objective/Outcome 2)
Key Outputs by Component (linked to the achievement of the Objective/Outcome 1)	1. Volume of SHS sold under the Project 2. Number of micro grids sold under the Project (Note: These outputs also support Objective/Outcome 2)
Objective/Outcome 2: Support the increased access to affordable electricity in the dispersed off-grid areas of Vanuatu	
Outcome Indicators	1. People provided with a new electricity service
Intermediate Results Indicators	1. Direct Project beneficiaries of which female 2. Total households of which are female headed households received a Solar Home System, micro grid or electricity through a mini grid 3. Participants in consultation activities during project implementation (number) 4. Participants in consultation activities during project implementation (female)
Key Outputs by Component (linked to the achievement of the Objective/Outcome 2)	1. Volume of SHS sold under the Project 2. Number of micro grids sold under the Project

**ANNEX 2. BANK LENDING AND IMPLEMENTATION SUPPORT/SUPERVISION****A. STAFF TIME AND COST**

Stage of Project Cycle	Staff Time and Cost	
	No. of staff weeks	US\$ (including travel and consultant costs)
Preparation		
FY17	22.547	85,545.03
FY18	0	0.00
FY19	0	850.02
FY20	0	1,800.32
Total	22.55	88,195.37
Supervision/ICR		
FY18	9.402	71,311.61
FY19	7.425	81,164.24
FY20	20.842	183,367.01
FY21	45.490	366,553.88
FY22	19.979	190,790.17
FY23	6.832	62,824.46
Total	109.97	956,011.37

**B. TASK TEAM MEMBERS**

Name	Role
Preparation	
Kamleshwar Prasad Khelawan	Task Team Leader(s)
Zhentu Liu	Procurement Specialist(s)
David Bruce Whitehead	Financial Management Specialist
Anuja Utz	Team Member
Rowena Margaret S. Gorospe	Counsel
Renee Berthome	Team Member
Kim Dagmar Baverstock	Team Member
Penelope Ruth Ferguson	Environmental Specialist
Ross James Butler	Social Specialist
Thimo Mueller	Team Member
Supervision/ICR	
Kamleshwar Prasad Khelawan, Mitsunori Motohashi, Yanqin Song, Toufiq Ahmed	Task Team Leader(s)
Michael Graeme Osborne	Procurement Specialist(s)
Janet Virginia Gamarra Rupa	Financial Management Specialist
Muzaffar Ahmad	Environmental Specialist
Fiona Evelyn Bingham	Team Member
Rebecca Dang	Team Member
Rebekah Beatrice Ramsay	Social Specialist
Nicholas Gerard Williams	Procurement Team
Nathalie Suzanna Noella Staelens	Environmental Specialist
Rachelle Therese Marburg	Social Specialist
Laisa Duri Boedoro	Procurement Team
Leisande Otto	Team Member
Dung Anh Hoang	Team Member
Enno Heijndermans	Team Member

**ANNEX 3. PROJECT COST BY COMPONENT**

Components	Amount at Approval (US\$M)	Actual at Project Closing (US\$M)	Percentage of Approval (US\$M)
Provision of Solar Home Systems and Micro Grids systems in rural areas of Vanuatu	5.37	0.30	14.33
Construction of Mini Grids systems in rural areas of Vanuatu	6.80	0	0
Technical Assistance and Project Management	2.00	1.70	73.50
Total	14.17	2.00	16.00



ANNEX 4. EFFICIENCY ANALYSIS

Not provided. No mini grids were procured or constructed, and no micro grids were purchased under the Project. An analysis was provided in the main text to cover the minimal volume of SHS purchases.



ANNEX 5. BORROWER, CO-FINANCIER AND OTHER PARTNER/STAKEHOLDER COMMENTS

The Borrower/Recipient provided comments that were incorporated directly into the main text of the ICR.



ANNEX 6. SUPPORTING DOCUMENTS (IF ANY)